

TUX IR

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Abstract

Our team **Tux IR** is developing two soccer robots for the RoboCup Junior Soccer 2v2 Infrared League.

The robots have a drive system with four omni-wheels powered by brushless motors, IR ball detection, line and distance sensors, and camera detection for goal scoring.

In the poster, we present **mechanics**, **electronics** and **software** and explain the most important design decisions and improvements made over the season.

Team & Overview

- **Team:** Tux IR, Germany.
- **League:** RoboCup Junior Soccer 2v2 Infrared.
- **Robots:** 2 identical robots for flexibility
- **Goals:** reliable line detection, stable and safe drive to the ball



Our Team

Mechanical design

- Compact, 3D-printed chassis for mounting the motors, kicker, and light sensors
- 4 omni-directional wheels for holonomic (360°) driving
- Three-stage structure: line detection and motors at the bottom, a central control board in the middle, additional sensors at the top

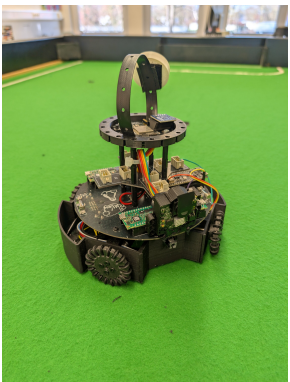
Development, Construction & Design

- Electronics**
- microcontroller: ESP32-S3
 - BLDC motor driver: Maxon ESCON 24/2
- Mechanics**
- CAD design in *Autodesk Fusion*.
 - Weight and drive system optimized to find a balance between agility and strength.
 - motors: GMP16T-TBC1636
- Sensors**
- long-range IR sensors from the CoRoSoN kit, minor software adjustments
 - near-field IR sensors made from 32 Vishay BPV10NF Photodiodes paired with LMV321 amplifiers for improved accuracy
 - Line detection with 32 phototransistors in a circular arrangement with WS2815 LEDs
 - CMPS14 compass provides precise directional values when the camera does not detect the goal.
- Software**
- Programmed in C++ via VS-Code + ESP IDF
 - Version history management: git
- Development effort**
- Development time: 6 months, usually 1-2 hours of team meetings per week
 - Work done mostly from home

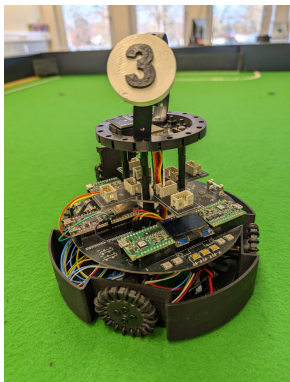
Bill of Materials (per Robot)

Component	Type	Quantity
Microcontroller	ESP32-S3	1
Motor driver	Maxon ESCON 24/2	4
BLDC-Gearmotor	GMP16T-TBC1636	4
Omni-Wheels	50mm, 3D-printed	4
long-range IR-Ballsensor	CoRoSoN-Kit IR-Ring	1
near-field IR-Ballsensor	Vishay BPV10NF	32
Linesensor	ALS-PT19-315C	32
IMU	CMPS14	1
Distance Sensors	DFRobot URM-09	3
Camera	Pixy 2.1	1

Images & Graphics



Front of Robot



Back of Robot

Software-architecture & Strategy

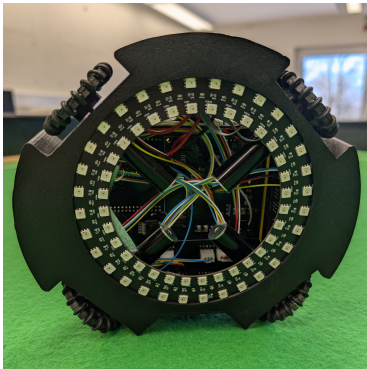
- Architecture**
- Low-Level: motor, sensor, and safety functions.
 - Mid-Level: data analysis (ball angle, compass, lines).
 - High-Level: game strategy.
- Strategy**
- Strategy: *Search, Angling towards goal*.
 - Priority: ball position → line avoidance → attack

Data, Results, & Discussion

- Data**
- controlled Vmax.: *0.5 m/s*.
 - line detection: *in 4ms, incorrect detection after crossing line only on ca. 1 Sensor*.
- Improvements**
- | characteristic value | before | after |
|------------------------|--------|-------|
| line detection [ms] | 6 | 4 |
| line misinterpretation | 7/20 | 3/20 |
- Discussion**
- Camera as goal detection allows facing towards the goal.
 - Driving according to vector directions enables high dynamics.
 - Limitations: Loss of ball when vision is obstructed, optimization for SuperTeam games planned.

Ball detection & line detection

- Ball detection (IR)**
- long-range Ring made up of 16 IR receivers and 32 near-field IR-Photodiodes, tuned to the modulation of the official IR ball.
 - Direction determination based on relative signal strengths.
 - Filtering using dynamic thresholds and time averaging.
- Line detection**
- 32 phototransistors and 64 LEDs near the edge of the robot.
 - When a line is detected, all directions of the active sensors towards the robot's center are added together to point back into the playing field.



Line-sensing PCB